Finitely Adequate Modules in Synthetic Algebraic Geometry

Thomas Thorbjørnsen | University of Western Ontario

Joint work with Felix Cherubini, Dan Christensen, Fabian Endres, and Hugo Moeneclaey

June 6, 2026

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Goal: Find a good internal generalization of externally finitely presented sheaves over the base ring.

Introduction

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Definition Let A be an fp algebra, and $a \in A$. The collection
$\mathrm{D}(a) :\equiv \{\phi \in \mathrm{Spec}(A) \mid \phi(a) \text{ is invertible}\} \subseteq \mathrm{Spec}(A)$ is a **basic open** of $\mathrm{Spec}(A)$.

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Axioms of SAG

1. **Locality**: For any $x, y \in R$ s.t. $x + y = 1$, either x is invertible or y is invertible.

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2. **Duality**: For any fp algebra A, the map $a \mapsto (\phi \mapsto \phi(a)) : A \to R^{\mathrm{Spec}(A)}$ is an R-algebra isomorphism.
3. **Zariski-Local Choice**: Let X be an affine scheme, and B an object. For any surjection $f : B \to X$, there exists a Zariski cover $\{U_i\}_{1,\ldots,n}$ of X with local sections $s_i : U_i \to B$ of f.

Definition An R-module P is a **finitely presented module (fp module)** if it is the cokernel of a map $f : R^m \to R^n$ for some natural numbers m and n. We denote the category of fp modules by $\mathrm{Mod}_R^{\mathrm{fp}}$.

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Theorem (CCEMT)
The category $\mathrm{Mod}_R^{\mathrm{fin.adeq.}}$ is extension closed in Mod_R.

Proof of Main Results

Definition An R-module L is **linearly adequate** if it is the kernel of a map $f : R^m \to R^n$ for some natural numbers m and n. We denote the category of linearly adequate modules by $\mathrm{Mod}_R^{\mathrm{lin.adeq.}}$.

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Theorem (Cherubini et al. [Che+25])
Duality $(-)^* :\equiv \mathrm{Hom}_R(-, R)$ induces an equivalence $\mathrm{Mod}_R^{\mathrm{fp}} \simeq (\mathrm{Mod}_R^{\text{lin.adeq.}})^{\mathrm{op}}$.

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Proposition (CCEMT)
M is finitely adequate if and only if it is the cokernel of a map $g : K \to L$, where K and L are linearly adequate.

Lemma (Key Lemma (to be proven), CCEMT)

Let L be linearly adequate and $e : E \twoheadrightarrow M$ a linear surjection with M finitely adequate. If either E or $\mathrm{Ker}(e)$ is finitely adequate, then any map $f : L \to M$ admits a linear lift $h : L \to E$.

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Proof.

($\Leftarrow$): By the Key Lemma.

$$\begin{tikzcd} & E \arrow[d, two heads, "e"] \\ L \arrow[ur, dashed] \arrow[r, "f"] & M \end{tikzcd}$$

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Corollary (CCEMT)

A finitely adequate module P is injective in $\mathrm{Mod}_R^{\text{fin.adeq}}$ if and only if it is finitely presented.

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Fact: By a result of Adelman [Ade73], finitely adequate modules form the **free abelian category** over finite free modules. Any additive functor F uniquely extends to an exact functor K

$$\begin{tikzcd} \mathrm{Mod}_R^{\text{fin.free}} \arrow[r, hook] \arrow[dr, "F"'] & \mathrm{Mod}_R^{\text{fin.adeq}} \arrow[d, dashed, "K"] \\ & \mathcal{A} \end{tikzcd}$$

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General case:

1. Pull back E along a presentation $p : L \twoheadrightarrow A$. The pullback p^*E is finitely adequate by the first case.

$$\begin{tikzcd} 0 \arrow[r] & B \arrow[r] \arrow[d, equal] & p^*E \arrow[r] \arrow[d, two heads] & L \arrow[r] \arrow[d, two heads, "p"] & 0 \\ 0 \arrow[r] & B \arrow[r] & E \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

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3. E is now finitely adequate, since it is the cokernel of the map $K \to p^*E$.

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Proof of the Key Lemma

Definition (The m-dimensional first-order infinitesimal disk)

$D(m) \equiv \mathrm{Spec}(R[x_1, \ldots, x_m]/\langle x_1, \ldots, x_m \rangle^2) \cong \{(\varepsilon_1, \ldots, \varepsilon_m) \in R^m \mid \varepsilon_i \varepsilon_j = 0 \text{ for all } i, j\}$

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Definition (Disk Inclusion) $\mathrm{inc}_{m,n} : D(m) \vee D(n) \to D(m+n)$ is defined as the wedge of maps $\varepsilon \in D(m) \mapsto (\varepsilon, 0)$ and $\varepsilon \in D(n) \mapsto (0, \varepsilon)$.

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Definition A pointed object (X, x) is **infinitesimally linear** if any diagram of pointed maps

$$\begin{tikzcd} D(m) \vee D(n) \arrow[r, "f"] \arrow[d, "\mathrm{inc}_{m,n}"] & X \\ D(m+n) \arrow[ur, dashed] & \end{tikzcd}$$

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Proposition (CCEMT)
Infinitesimally linear R-modules at 0 are closed under finite limits, finite colimits, and extensions.

Example Finitely adequate modules are infinitesimally linear at 0.

Definition The **tangent space** of a pointed object (X, x) is the object of pointed maps $T_x X := (X, x)^{(D(1),0)}$.

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The tangent space T_0M has an R-module structure given by pointwise addition and scalar multiplication.

We define **the canonical map** $\mathrm{can} : M \to T_0M$ by $m \mapsto (\varepsilon \mapsto \varepsilon \cdot m)$.

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Example Finitely adequate modules are tangent-like.

Theorem (CCEMT)
Let $f : (M, 0) \to (N, 0)$ be a pointed map of tangent-like R-modules. The derivative df_0 is linear on the tangent spaces.

Theorem (Linearization Trick, CCEMT)

Suppose we have a commuting square of R-modules where all the modules are tangent-like. If there exists a pointed lift $h : B \to X$, then there exists a linear lift.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "m"] & X \arrow[d, "e"] \\ B \arrow[r, "g"] \arrow[ur, dotted, "h"] & Y \end{tikzcd}$$

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Proof. Since can is a natural equivalence on all objects in the square, it is sufficient to find a linear lift on the tangent spaces. The derivative dh_0 is such a linear lift. $\square$

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Lemma (Key Lemma, CCEMT)

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